Scuola universitaria professionale della Svizzera italiana
Dipartimento tecnologie innovative
Istituto sistemi informativi e networking

SUPSI

Introduction

Operating Environments

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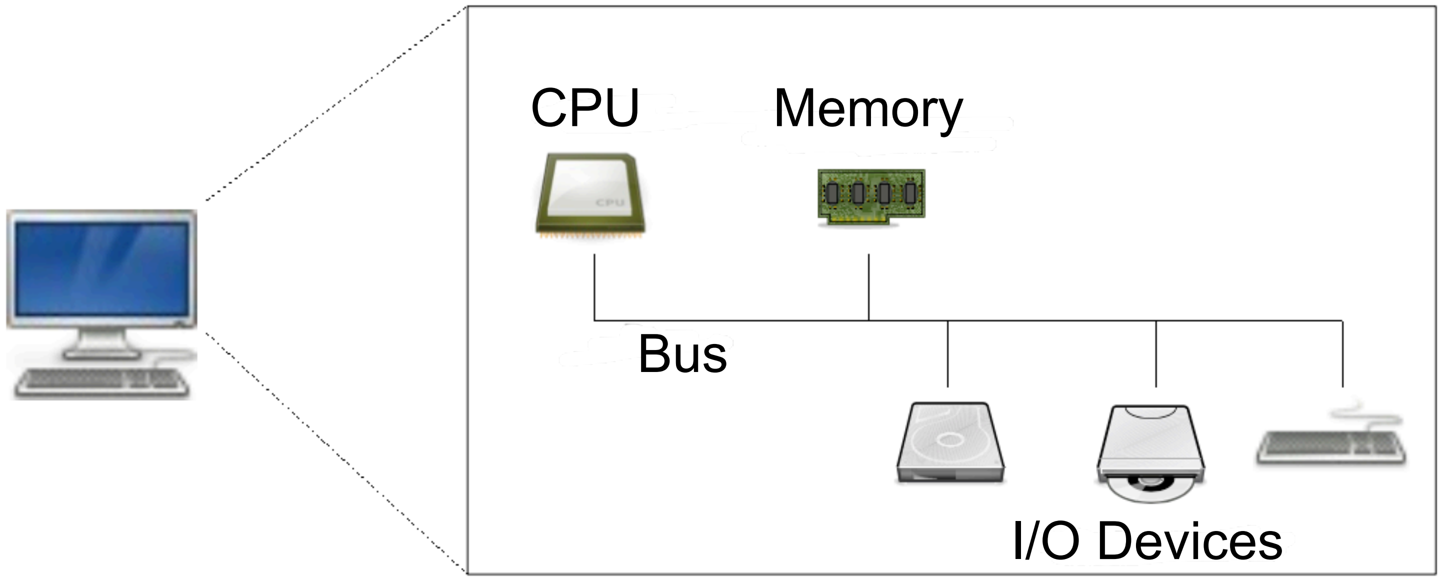
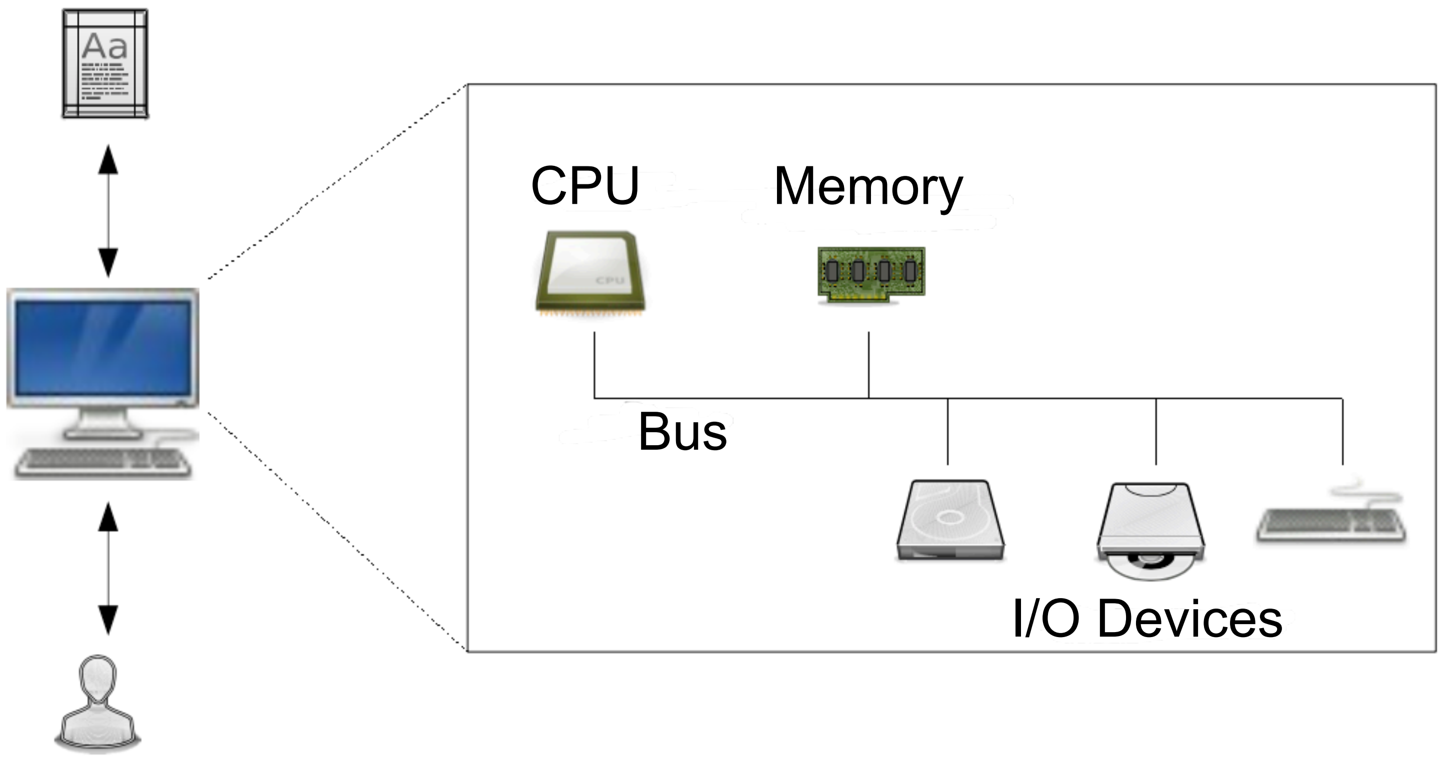
Lesson's Goals

- Study the basic principles of a computer.
- Understand the functionality of an operating system.
- Understand the difference between an operating system and an operating environment.

The computer and its architecture



Software execution



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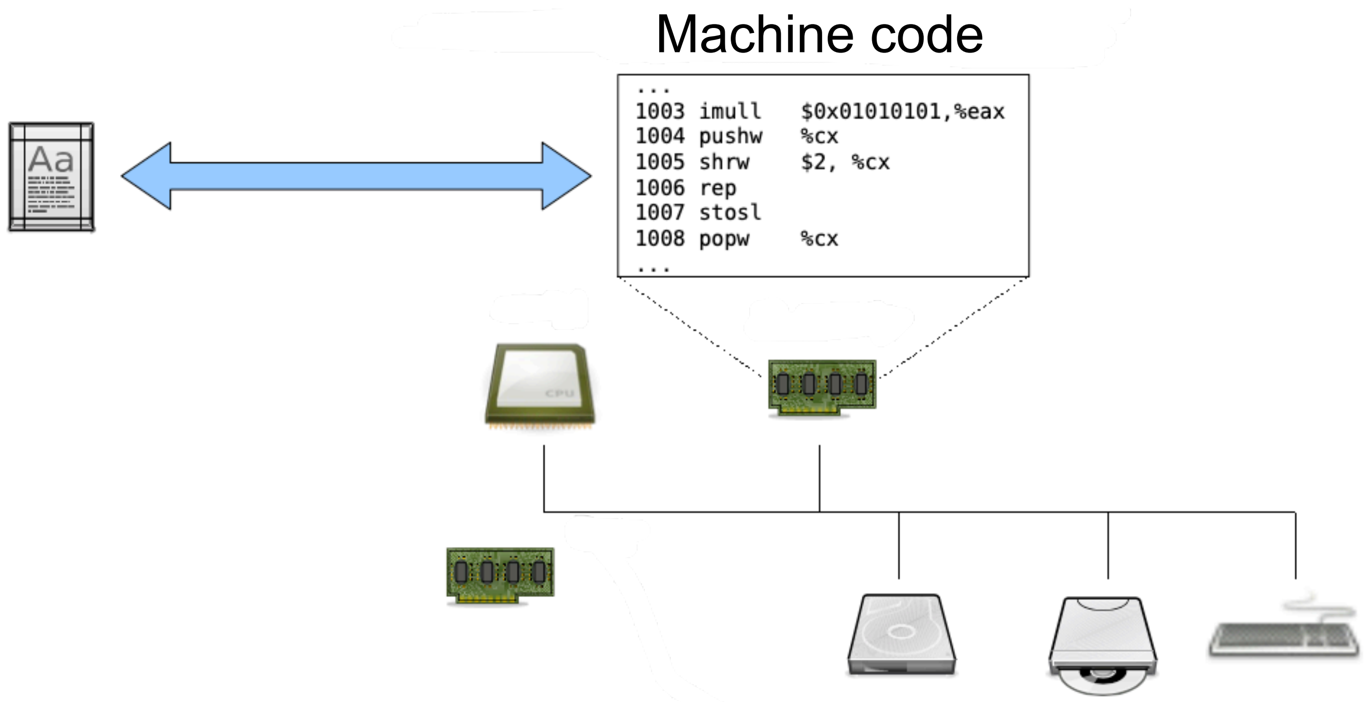
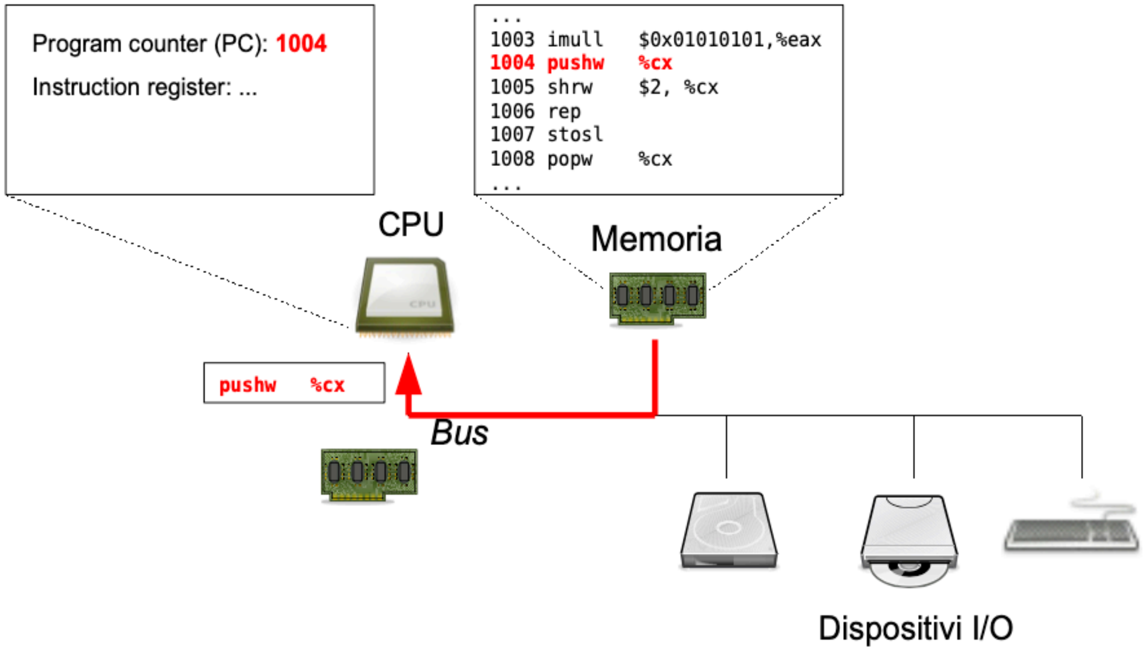
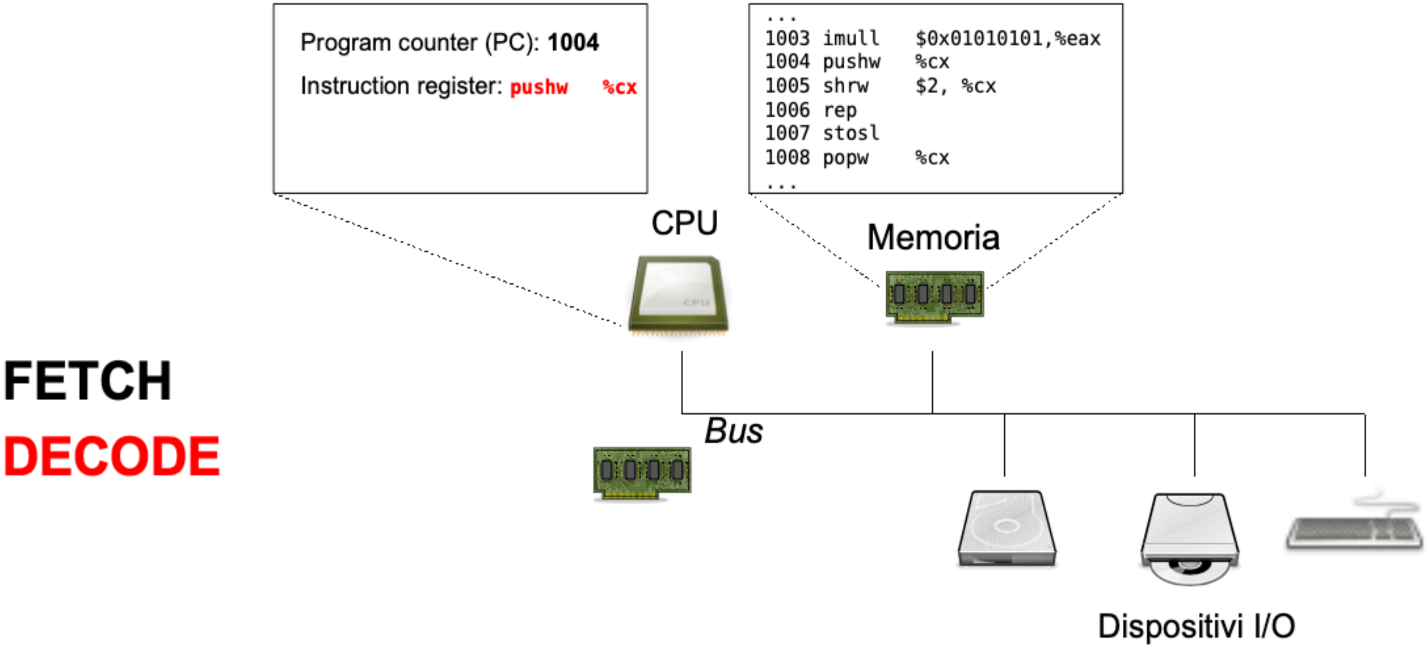


Software execution

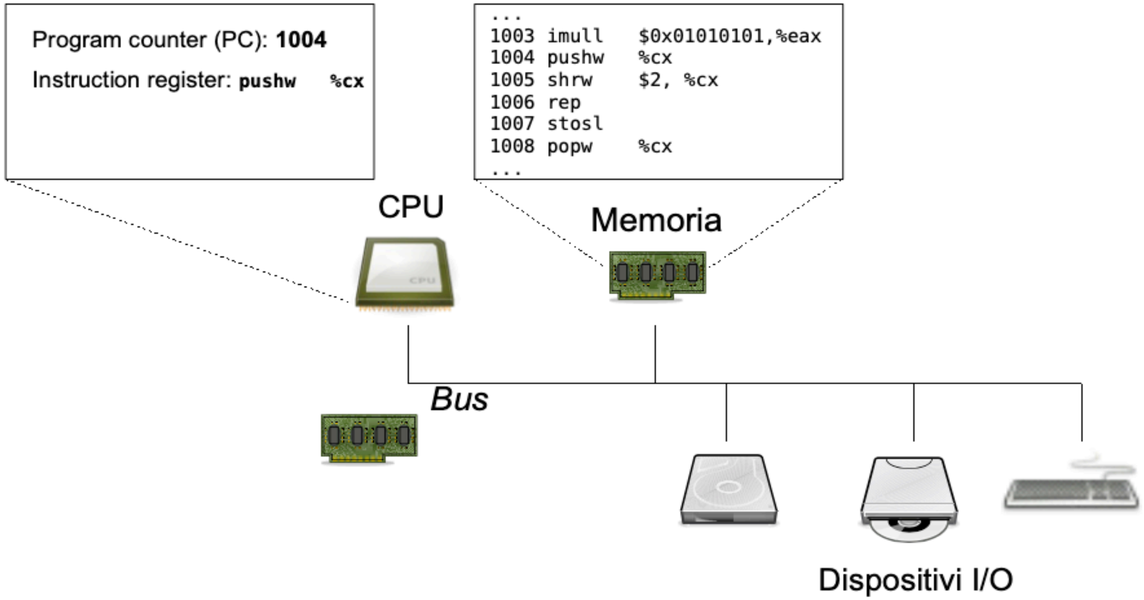
FETCH



Software execution

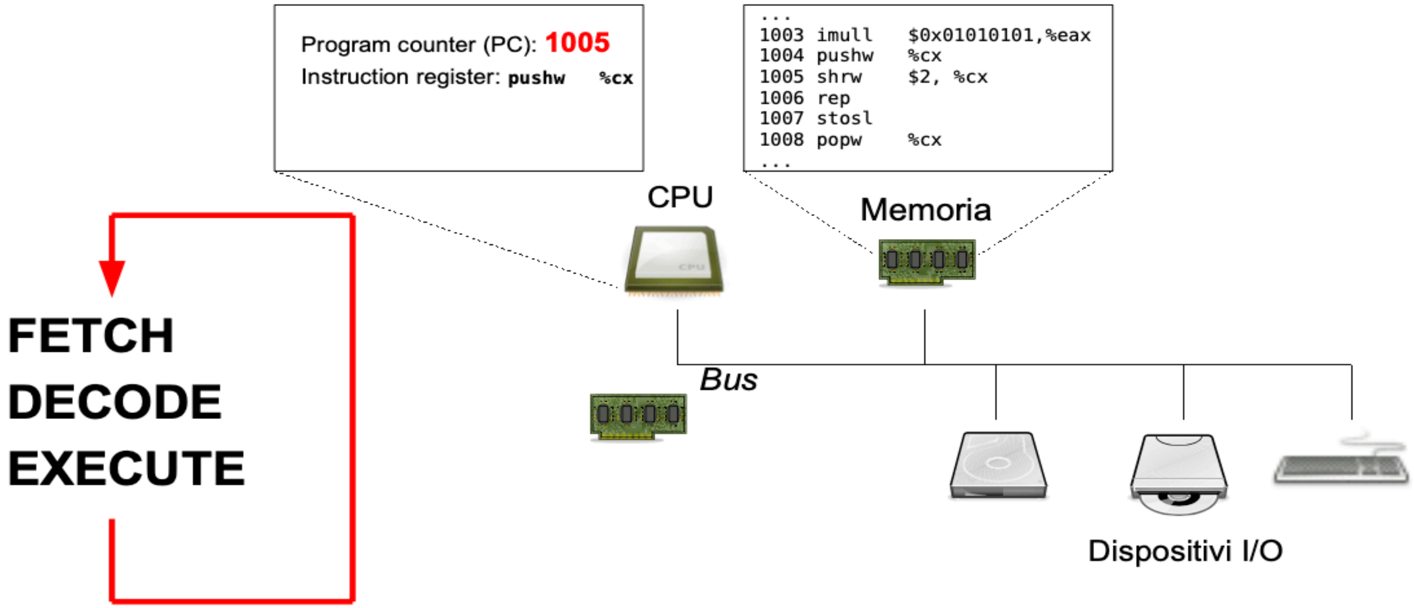


Software execution



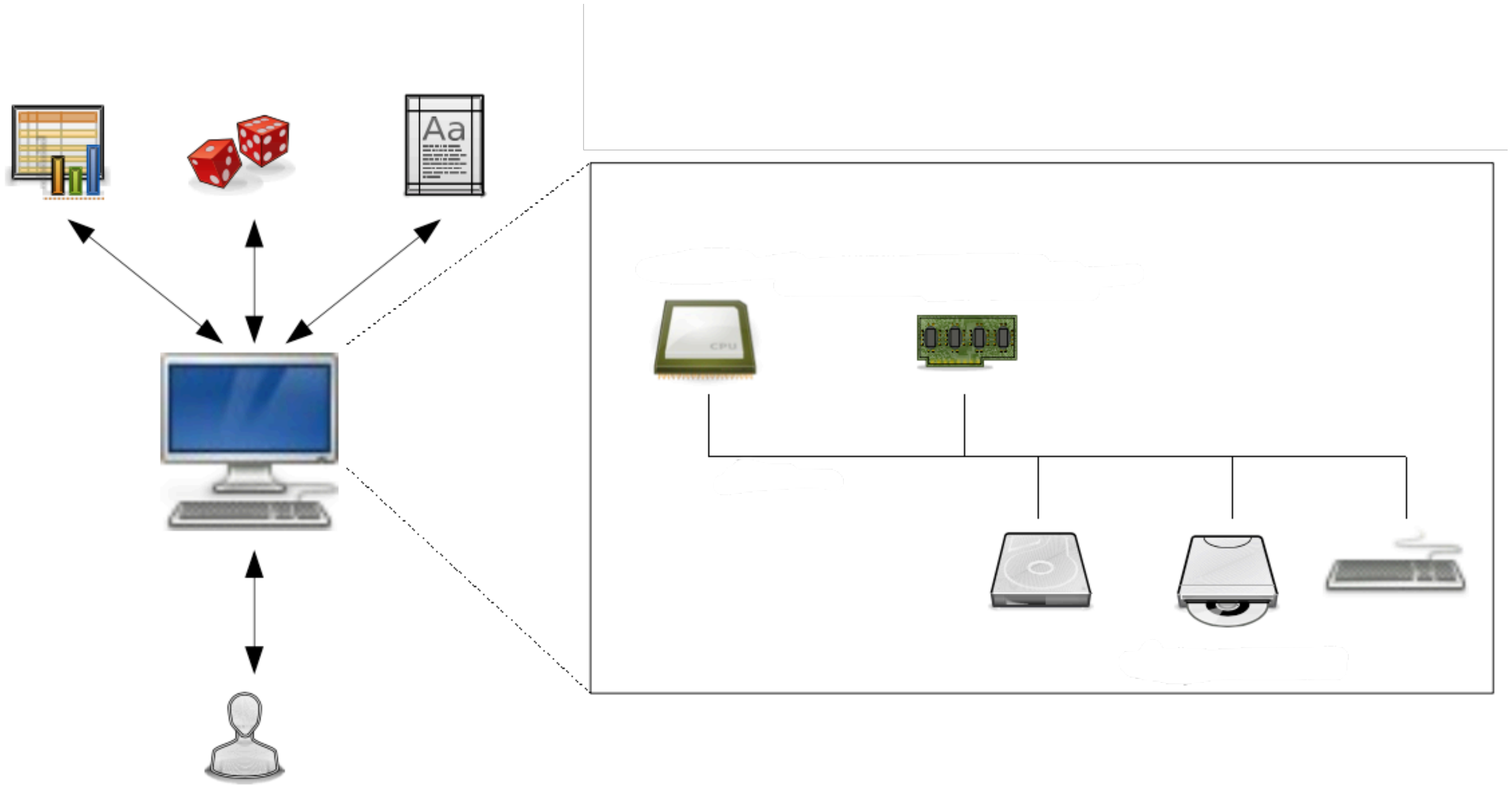
FETCH
DECODE
EXECUTE

Software execution



Multi process system

- Hardware differences?
- File access?
- Access to resources?
- Concurrency?



Multi process, multi user system

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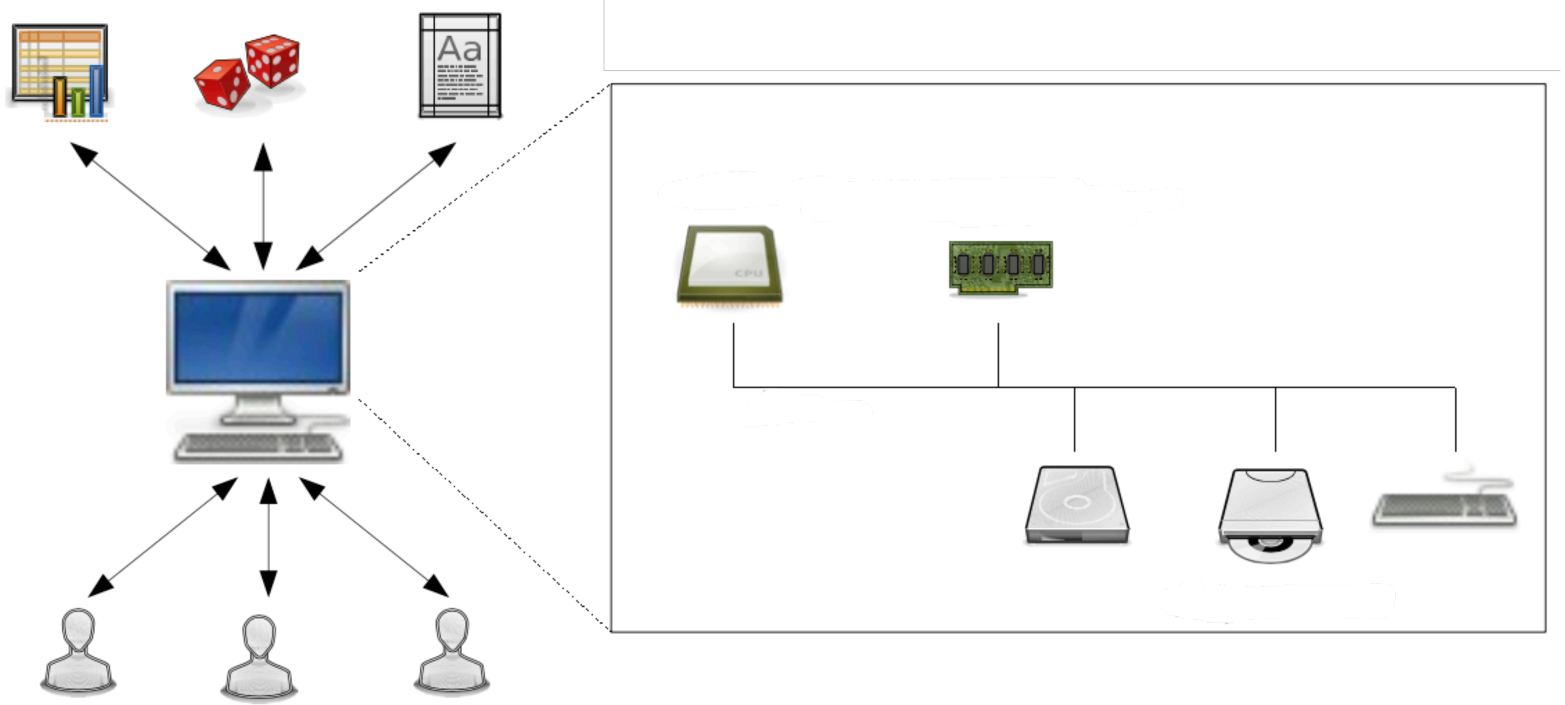
- Concurrency?
- Files organisation?
- Programs execution?
- Device use?
- Security?



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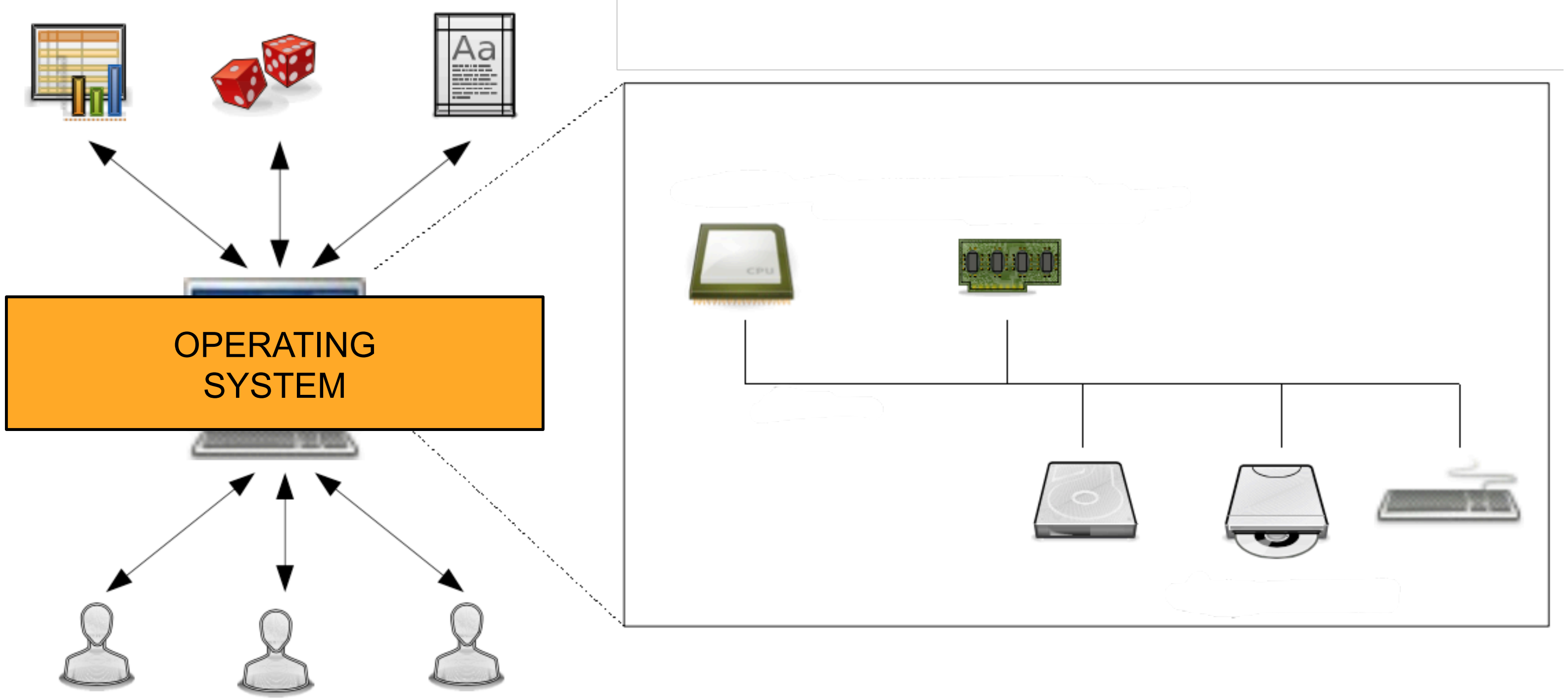
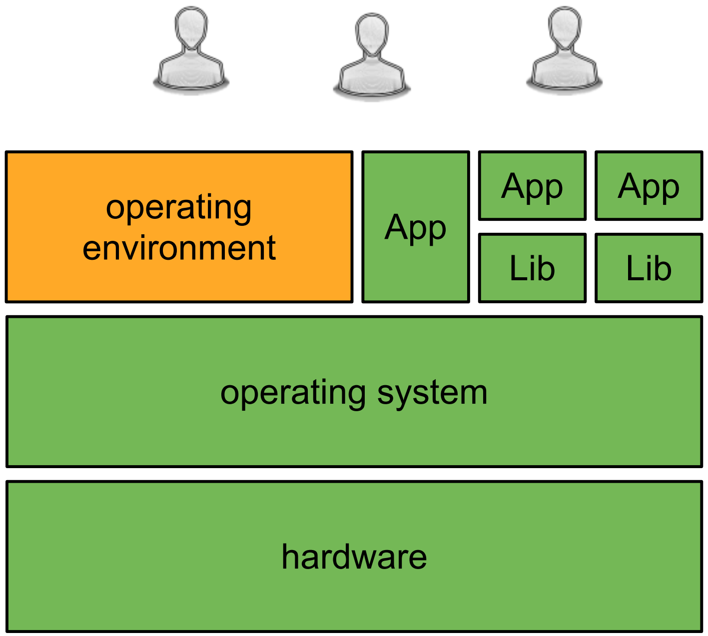


What is an operating system?

- **Abstraction** to hide the management mechanisms of the machine
  - Simplifies the programming and the use: the user is able to use the physical machine without knowing the details of its internal structure and its operation.

- **Resource Manager**
  - Allows all resources to work efficiently together and provides a consistent and comprehensive view of the computer.

Above and below an operating system



users

Applications: allow the user to solve specific problems
Libraries: group common features that can be reused between different applications
Operating environment

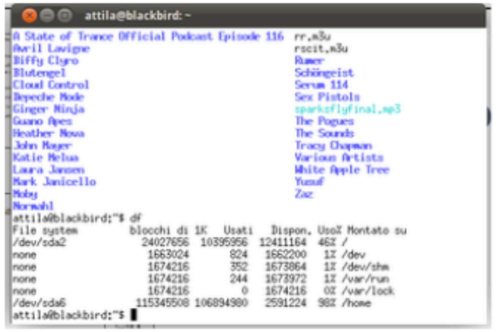
Operating system (kernel, driver): controls and coordinates the use of hardware between applications and users active on the system

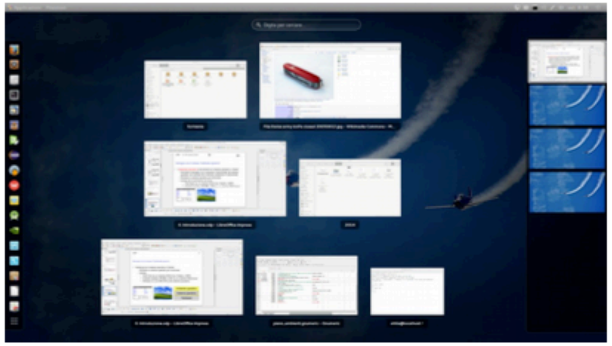
Hardware: provides computing and storage resources (CPU, memory, I/O peripherals)

Interact with the system: the operating environment

- The operating environment acts as a **link between the operating system and the user**:
  - Allows the user to interact with the abstractions implemented by the operating system (e.g. files and directories, running programs).
  - Requires an operating system to run

- Examples that we will meet during the course:
  - The Bash shell
  - Desktop environments (like GNOME and KDE)





How to get to the operating environment?